

ECG Arrhythmia Detector

A MATLAB-Based Prototype for Signal Analysis and Classification

The Problem & Project Objectives

The Problem

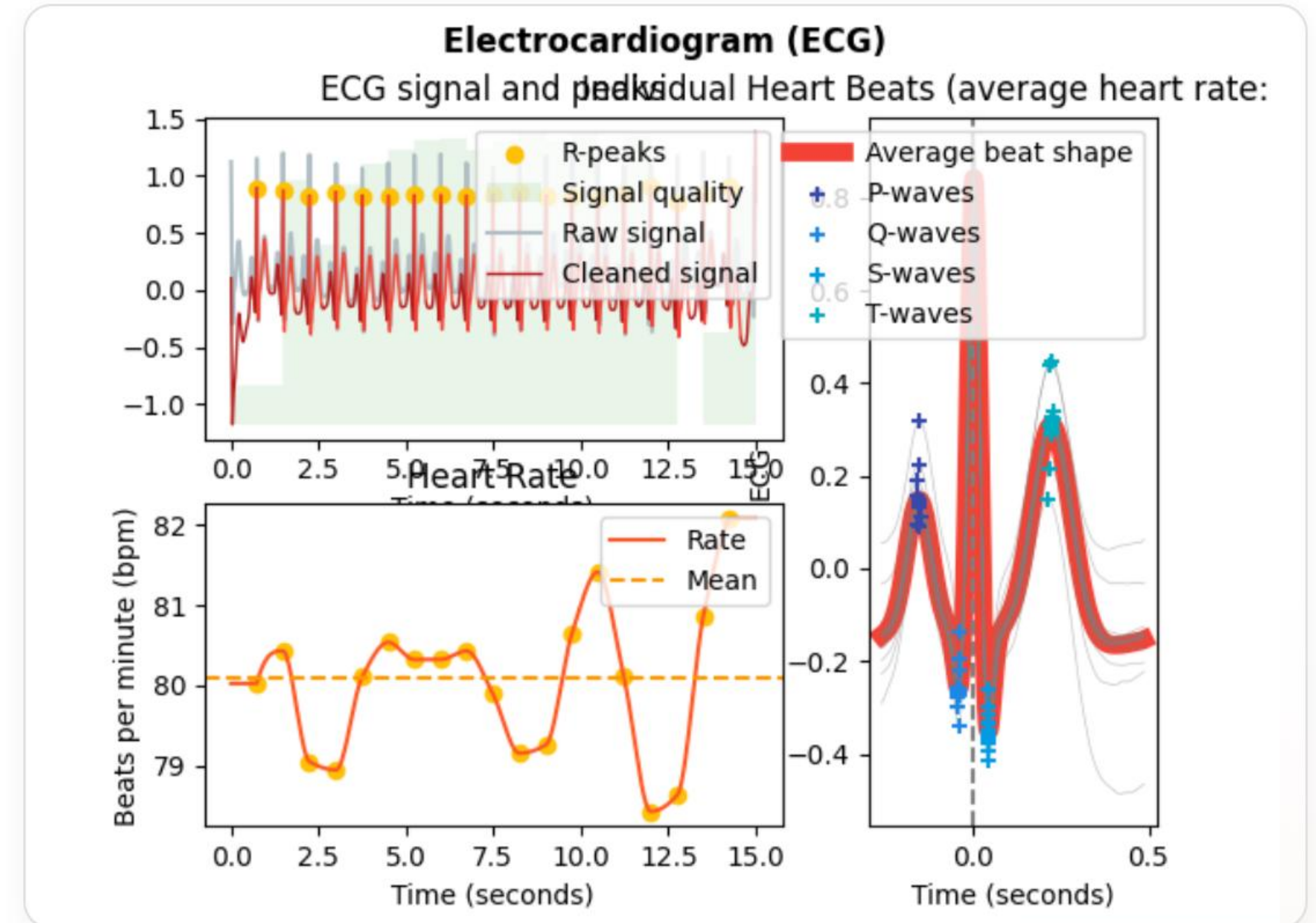
Raw ECG signals are unusable, corrupted by multiple sources of noise:

- Baseline Wander (from breathing)
- Powerline Hum (50/60 Hz interference)
- Muscle (EMG) Noise (high-frequency)

The Objective

To design a DSP prototype in MATLAB to:

- Clean and filter a raw ECG signal.
- Automatically detect each heartbeat (QRS complex).
- Classify the heart rate (Normal, Tachycardia, Bradycardia).

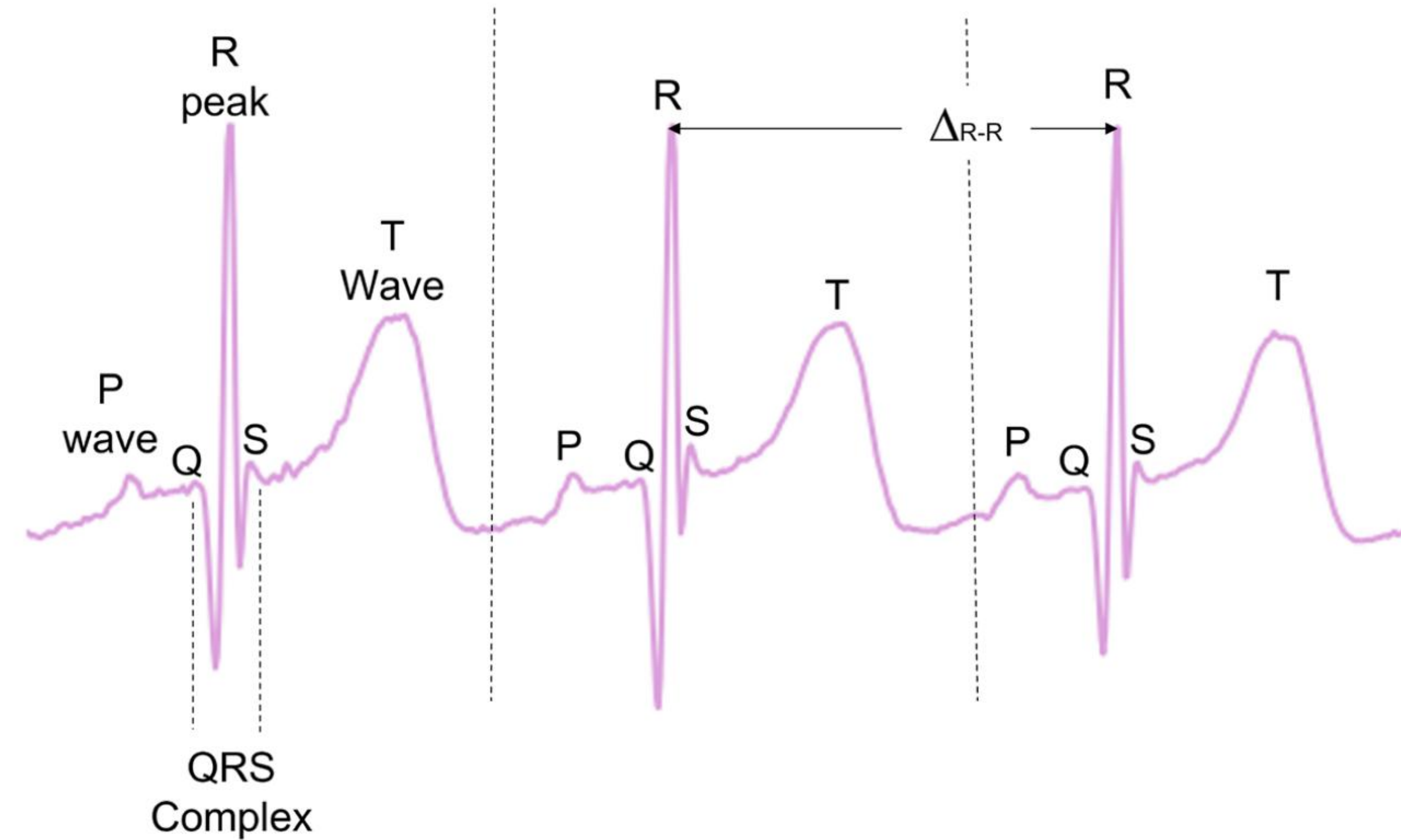


Raw ECG signal is obscured by noise.

Step 1: Signal Pre-processing

A three-stage filter cascade was designed to isolate the clean ECG signal.

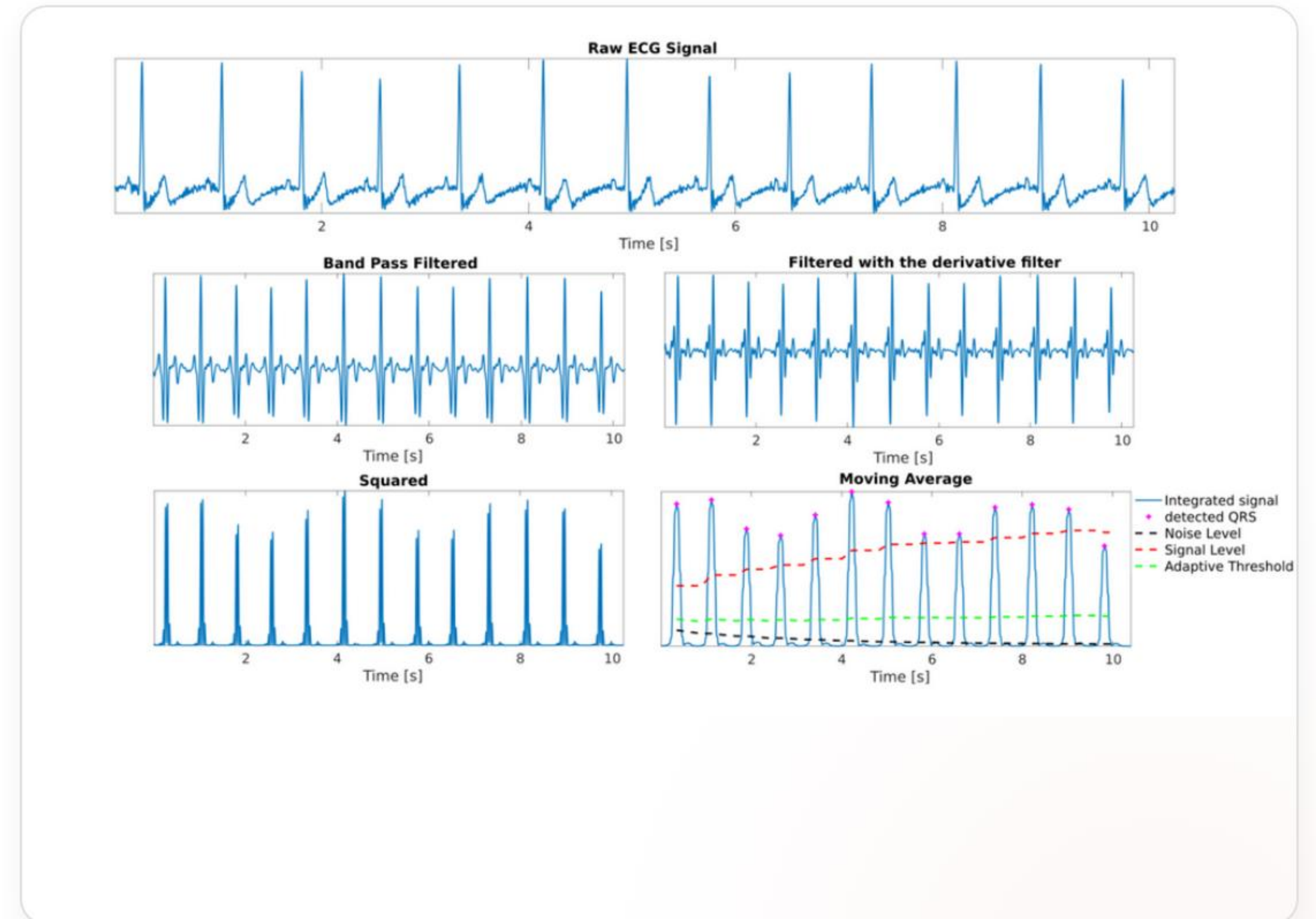
-  **High-Pass Filter**
Removes low-frequency Baseline Wander.
-  **Notch Filter**
Targets and removes 60 Hz powerline interference.
-  **Low-Pass Filter**
Smooths the signal by removing high-frequency muscle noise.



Step 2: QRS Detection (Pan-Tompkins)

To find the heartbeat, we used a method based on the Pan-Tompkins algorithm:

- > **Derivative:** The filtered signal is differentiated to find the steep slopes of the R-wave.
- > **Squaring:** This enhances the derivative's output and makes all points positive.
- > **Moving-Window Integration:** A smoothing step that creates a clear "feature pulse" for each heartbeat.
- > **FindPeaks:** An adaptive threshold is applied to this integrated signal to find the exact location of each beat.

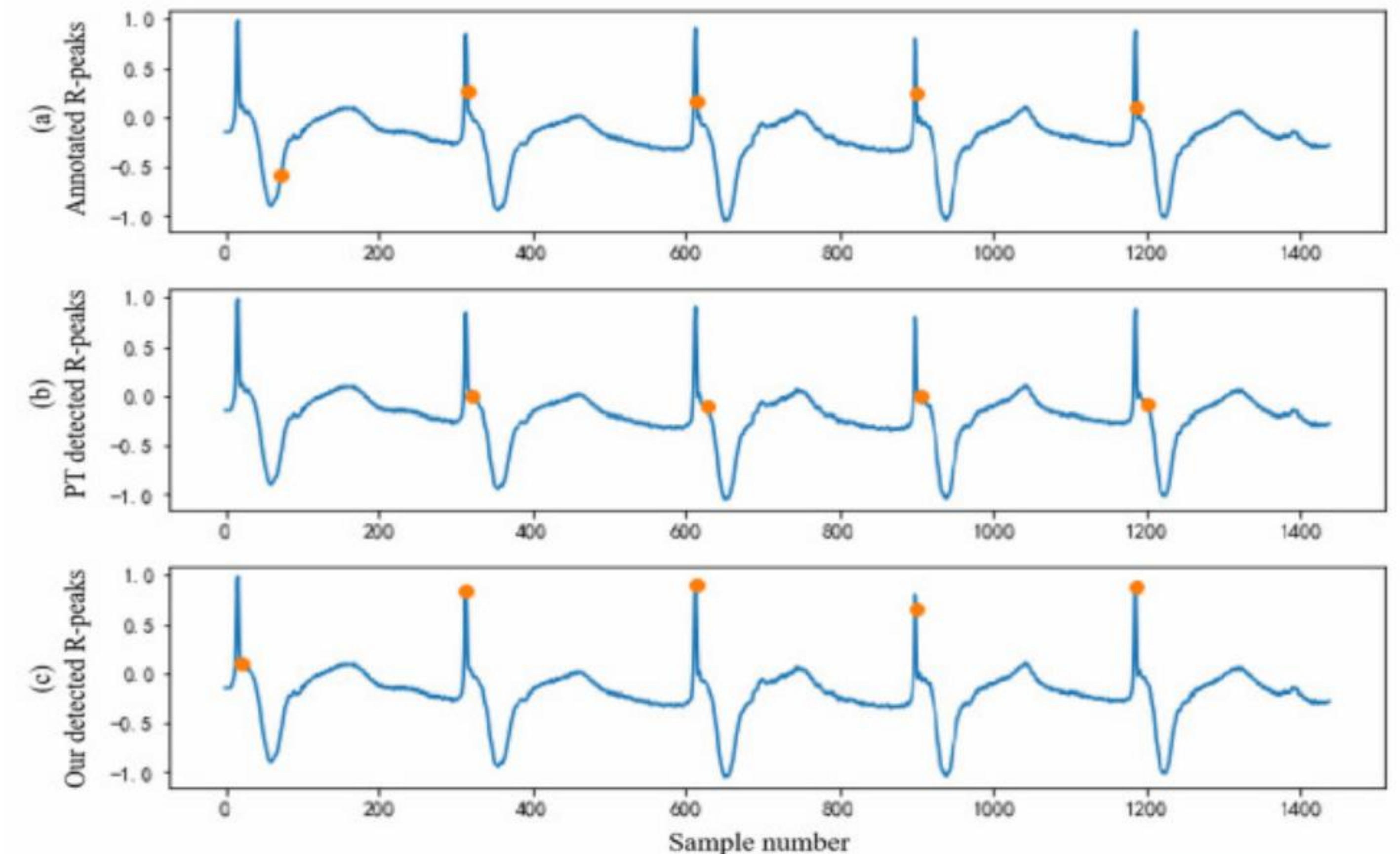


The integrated signal (bottom) creates clear peaks for detection.

Step 3: Analysis & Results

With the R-peaks located, we can perform the final analysis:

- > **R-R Interval:** The time (or number of samples) between consecutive R-peaks is calculated.
- > **BPM Calculation:** The R-R interval is converted into instantaneous Beats Per Minute (BPM).
- > **Classification:** Each BPM value is classified based on clinical thresholds:
 - > **Tachycardia:** > 100 BPM
 - > **Normal:** 60 - 100 BPM
 - > **Bradycardia:** < 60 BPM



Final output: Detected R-peaks (red dots) on the filtered signal.

Conclusion & Future Work

This project successfully created a MATLAB prototype that can load, filter, and analyze an ECG signal to classify arrhythmias.

Future work includes porting this algorithm to a real-time DSP and expanding detection for more complex arrhythmias.

Thank You

Questions?